```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
data = load_breast_cancer()

X = data.data
y = data.target

# Convert to DataFrame (optional but useful)
df = pd.DataFrame(X, columns=data.feature_names)
df['target'] = y

df.head()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67

5 rows × 31 columns

```
print("Dataset Shape:", df.shape)

print("\nTarget Classes:", data.target_names)

print("\nClass Distribution:")
print(df['target'].value_counts())
```

```
Dataset Shape: (569, 31)

Target Classes: ['malignant' 'benign']

Class Distribution:
target
1    357
0    212
Name: count, dtype: int64
```

```
X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42,
    stratify=y
)
```

```
knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)
```

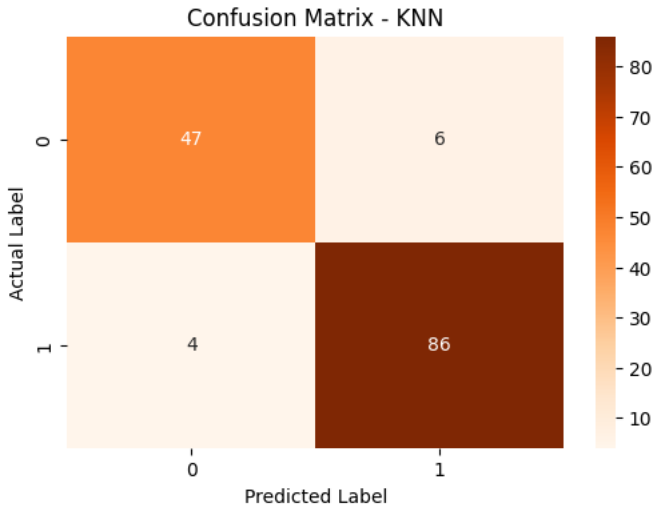
```
▾ KNeighborsClassifier
KNeighborsClassifier()
```

```
y_pred_knn = knn_model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred_knn)

print("KNN Model Accuracy:", accuracy)
```

```
KNN Model Accuracy: 0.9300699300699301
```

```
cm = confusion_matrix(y_test, y_pred_knn)

plt.figure(figsize=(6,4))

sns.heatmap(cm, annot=True, fmt='d', cmap='Oranges')

plt.title("Confusion Matrix - KNN")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")

plt.show()
```



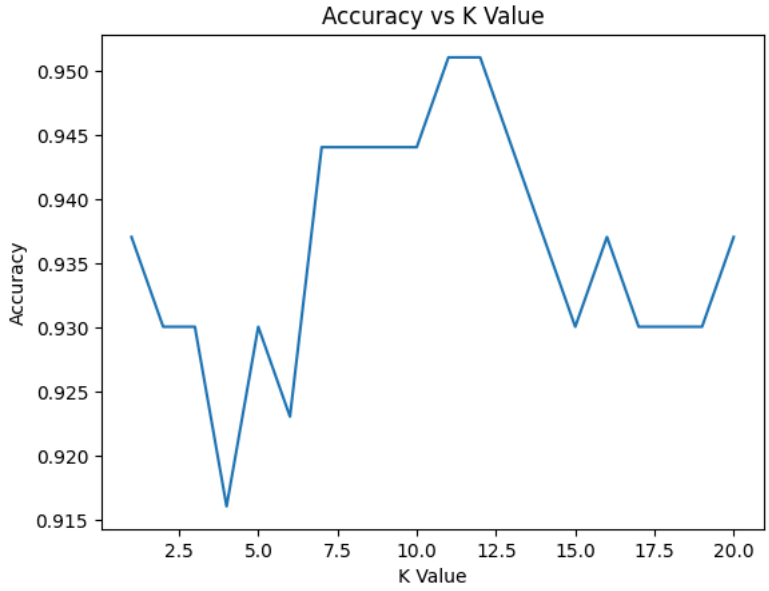

```
print("Classification Report:\n")

print(classification_report(y_test, y_pred_knn))
```

```
Classification Report:

              precision    recall  f1-score   support

           0       0.92      0.89      0.90        53
           1       0.93      0.96      0.95        90

    accuracy                           0.93       143
   macro avg       0.93      0.92      0.92       143
weighted avg       0.93      0.93      0.93       143
```

```
knn_accuracy = accuracy_score(y_test, y_pred_knn)

print("KNN Accuracy:", knn_accuracy)
```

```
KNN Accuracy: 0.9300699300699301
```

```
accuracy_values = []

for k in range(1,21):

    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)

    pred = model.predict(X_test)

    accuracy_values.append(accuracy_score(y_test, pred))

plt.plot(range(1,21), accuracy_values)

plt.xlabel("K Value")
plt.ylabel("Accuracy")
plt.title("Accuracy vs K Value")

plt.show()
```




```
print("KNN Classification Report:\n")
print(classification_report(y_test, y_pred_knn))
```

```
KNN Classification Report:

              precision    recall  f1-score   support

           0       0.92      0.89      0.90        53
           1       0.93      0.96      0.95        90

    accuracy                           0.93       143
   macro avg       0.93      0.92      0.92       143
weighted avg       0.93      0.93      0.93       143
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import precision_recall_curve, roc_curve, auc
```

```
svm_model = SVC(kernel='linear', probability=True)

svm_model.fit(X_train, y_train)
```

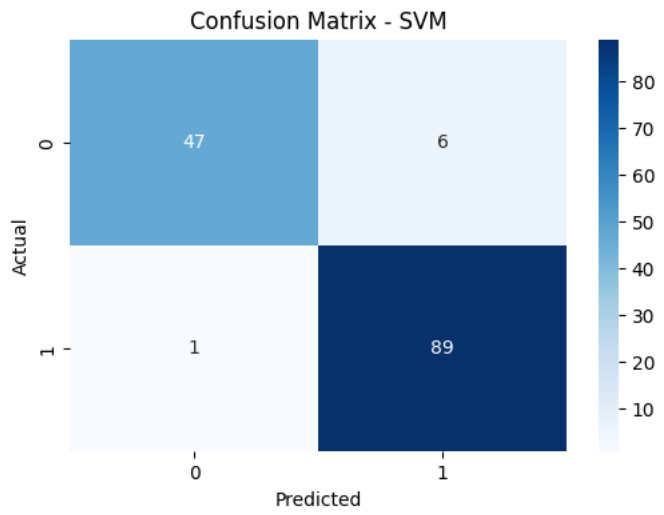
```
▾ SVC
SVC(kernel='linear', probability=True)
```

```
y_pred_svm = svm_model.predict(X_test)
```

```
svm_accuracy = accuracy_score(y_test, y_pred_svm)

print("SVM Accuracy:", svm_accuracy)
```

```
SVM Accuracy: 0.951048951048951
```

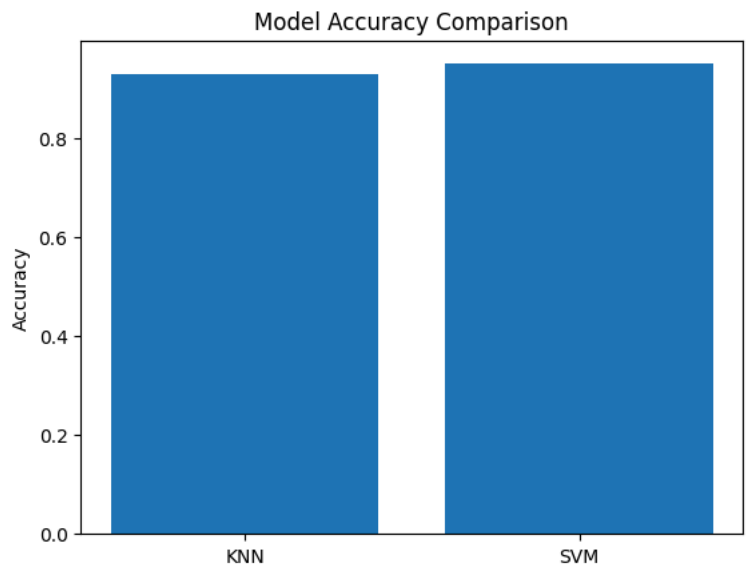
```
cm_svm = confusion_matrix(y_test, y_pred_svm)

plt.figure(figsize=(6,4))
sns.heatmap(cm_svm, annot=True, fmt='d', cmap='Blues')

plt.title("Confusion Matrix - SVM")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()
```



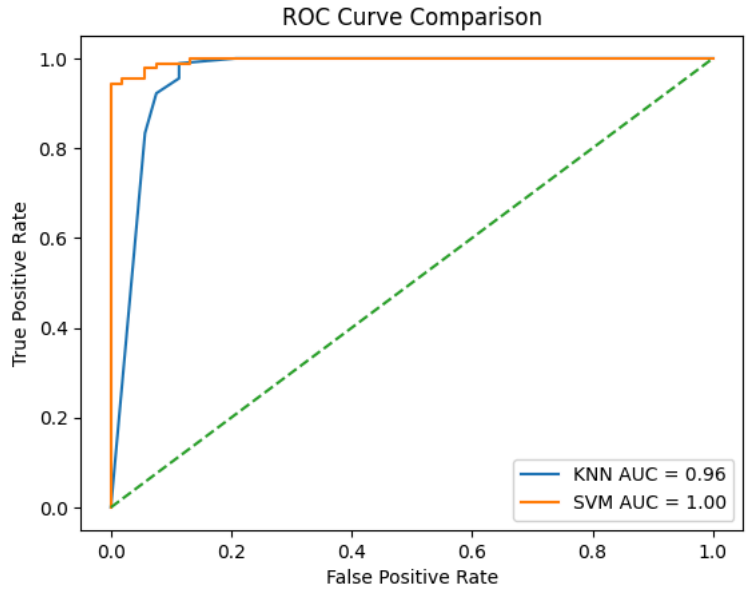

```
print("SVM Classification Report:\n")
print(classification_report(y_test, y_pred_svm))
```

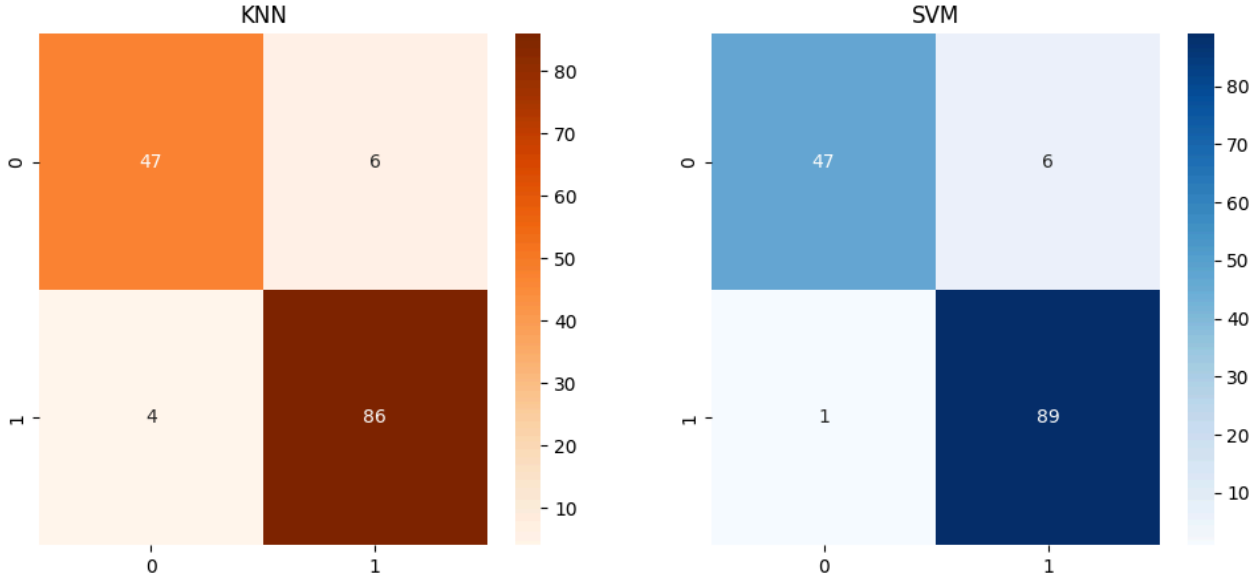
```
SVM Classification Report:

              precision    recall  f1-score   support

           0       0.98      0.89      0.93        53
           1       0.94      0.99      0.96        90

    accuracy                           0.95       143
   macro avg       0.96      0.94      0.95       143
weighted avg       0.95      0.95      0.95       143
```

```
models = ['KNN', 'SVM']
accuracies = [knn_accuracy, svm_accuracy]

plt.bar(models, accuracies)

plt.ylabel("Accuracy")
plt.title("Model Accuracy Comparison")

plt.show()
```



```
knn_prob = knn_model.predict_proba(X_test)[:,1]
svm_prob = svm_model.predict_proba(X_test)[:,1]

fpr_knn, tpr_knn, _ = roc_curve(y_test, knn_prob)
fpr_svm, tpr_svm, _ = roc_curve(y_test, svm_prob)

auc_knn = auc(fpr_knn, tpr_knn)
auc_svm = auc(fpr_svm, tpr_svm)

plt.plot(fpr_knn, tpr_knn, label="KNN AUC = %.2f" % auc_knn)
plt.plot(fpr_svm, tpr_svm, label="SVM AUC = %.2f" % auc_svm)

plt.plot([0,1],[0,1],'--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")

plt.legend()

plt.show()
```




```
fig, ax = plt.subplots(1,2, figsize=(12,5))

sns.heatmap(cm_knn, annot=True, fmt='d', cmap='Oranges', ax=ax[0])
ax[0].set_title("KNN")

sns.heatmap(cm_svm, annot=True, fmt='d', cmap='Blues', ax=ax[1])
ax[1].set_title("SVM")

plt.show()
```